

The diagram illustrates the pin configuration for the SH76E000 microcontroller. It features two rows of pins, each with its corresponding function listed next to it.

Pin	Function
SH3	
HV_RXD	HV_RX
HV_TXD	HV_TX
HV_NRST	HV_NRST
HV_SWDIO	HV_SWDIO
HV_SWDCLK	HV_SWDCLK
Mon_SLK_buf	Mon_SLK_buf
Mon_LMP_buf	Mon_LMP_buf
Mon_IL_prot	Mon_IL_prot
SUP_RXD	SUP_RX
SUP_TXD	SUP_TX
HV_EXT_EN	HV_EXT_EN
LMP_EN	LMP_EN
Light_sns	Light_sns
Vbus_ctrl	Vbus_ctrl
SH0	

The schematic diagram illustrates the electrical connections for the STM32G431K8Tx microcontroller (U60). The microcontroller is shown in a yellow box with pins 1 through 25 on the right side and pins 16, 14, and 15 on the bottom. The following components and connections are detailed:

- Power and Grounding:**
 - VDD (Pin 1):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - VDDA (Pin 15):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - VSS (Pin 16):** Connected to ground (SH0).
 - VSSA (Pin 14):** Connected to ground (SH0).
- Reset and Control:**
 - PGIO (Pin 4):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - PF0 (Pin 2):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - PF1 (Pin 3):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - BOOT0 (Pin 31):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - BOOT1 (Pin 30):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
- External Signals and Test Points:**
 - TP60 (Pin 5):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP61 (Pin 6):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP62 (Pin 9):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP64 (Pin 10):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP66 (Pin 11):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP68 (Pin 12):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP70 (Pin 19):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP71 (Pin 20):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP72 (Pin 21):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP74 (Pin 25):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP75 (Pin 16):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP76 (Pin 14):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
 - TP77 (Pin 15):** Connected to a network of capacitors (C60, C61, C62, C63) and a resistor (R62) to ground (SH0).
- Other Components:**
 - Y60 (20 MHz):** A crystal connected to pins 13 (PB0) and 14 (PB1).
 - R62 (10k):** A resistor connected to pins 1, 4, 15, and 31.
 - R64 (10k):** A resistor connected to pins 30 and 31.
 - C60, C61, C62, C63 (100n):** Capacitors connected to pins 1, 4, 15, and 31.
 - C64 (100n):** A capacitor connected to pin 30.
 - C65 (22p):** A capacitor connected to pin 13.
 - C66 (22p):** A capacitor connected to pin 14.

The schematic diagram illustrates the LED driver circuit. It consists of two LEDs, D60 and D61, connected in series. Resistor R63 (1k) is connected in series with the LEDs, and resistor R65 (10k) is connected in series with the ground. The circuit is powered by SH3 and grounded at SH0. A control signal ST_LED is connected to the junction between R63 and R65. Diodes D60 and D61 are marked with a red arrow indicating they are to be replaced with LEDs.



MCU

/Block Diagram/MCU/

Main board

MCU.kicad_sch

CHECKED

1 kV strobe

2025-07-04

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